



RARB Natural History Study

Valerie Chu, MSc, CGC

Genetic Counsellor and Research Coordinator



Université  de Montréal



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Why?

- Variation among affected individuals
- Different forms and courses of this condition poorly understood
- Not knowing how a disease typically manifests, or what symptoms may appear, disappear, or progress, makes it difficult to develop and assess therapies
- Better understand how the RARB-related disorder manifests and evolves over a period of time

A Natural History Study Can...

- Provide a comprehensive understanding of *RARB*-related disorders and its clinical course in an individual's lifetime
- Identify patients' and families' unmet medical needs
- Improve diagnosis of this disorder
- Establish standardized clinical care guidelines
- Identify potential targets or biomarkers
- Inform clinical trials



Research Team

- **Jacques Michaud**, Principal Investigator
- **Inge Meijer**, Neurologist
- **Sarah Lippé**, Neuropsychologist
- **André Tremblay**, Scientist
- **Valerie Chu**, Study Coordinator



Executive Committee

- **Reinhard Pell**, Cure MCOPS12
- **Nina** and **Anthony Flint**, A Cure for Sienna
- **Rachel Sutherland**, A Cure for Sophia and Friends

Eligibility Criteria

Any age, any country, with pathogenic, likely pathogenic variant(s), or VUS in *RARB*

Non-English and non-French speakers eligible

Fetal and deceased cases eligible

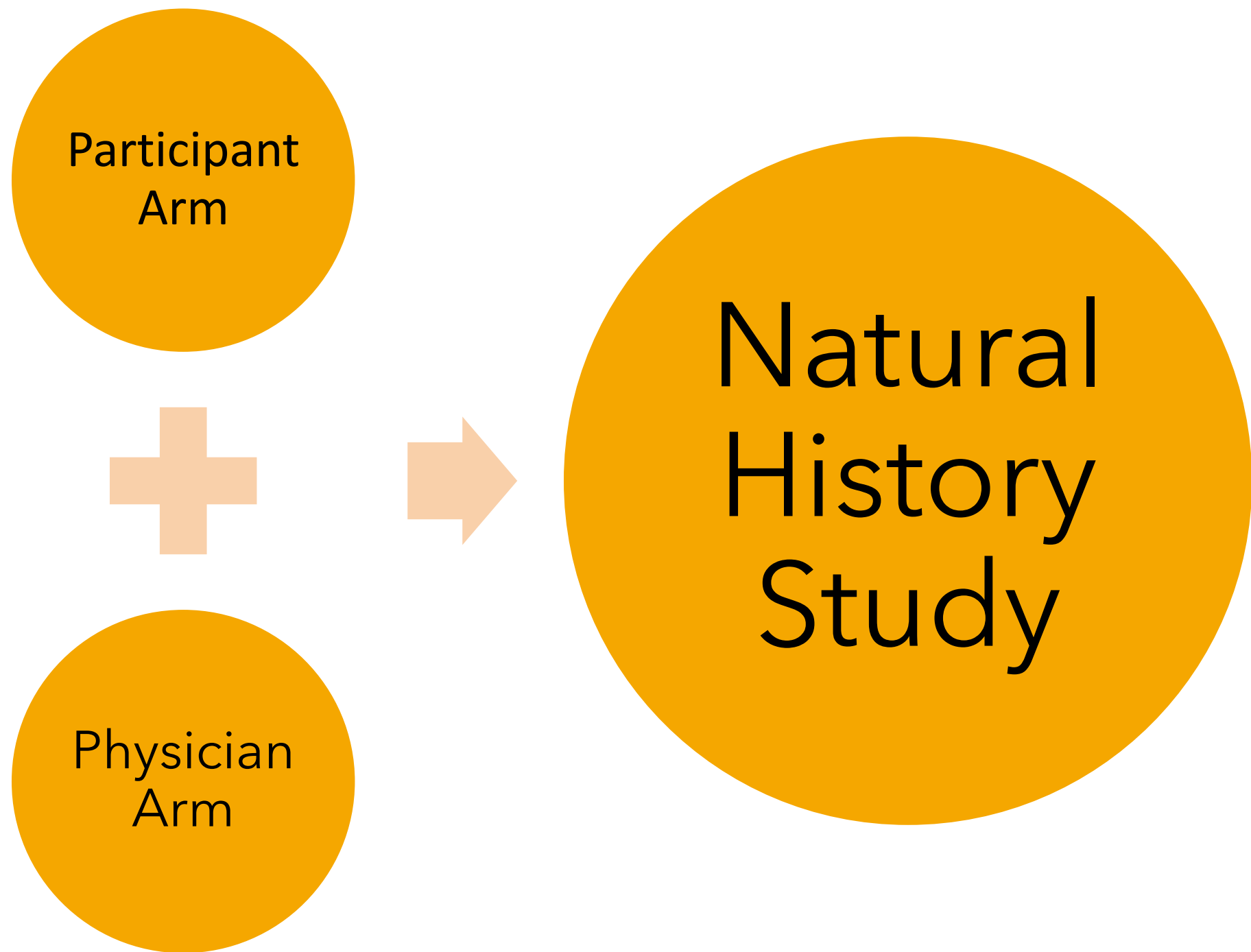


With study modifications



E-Consenting and Database





Vineland interview

Annual

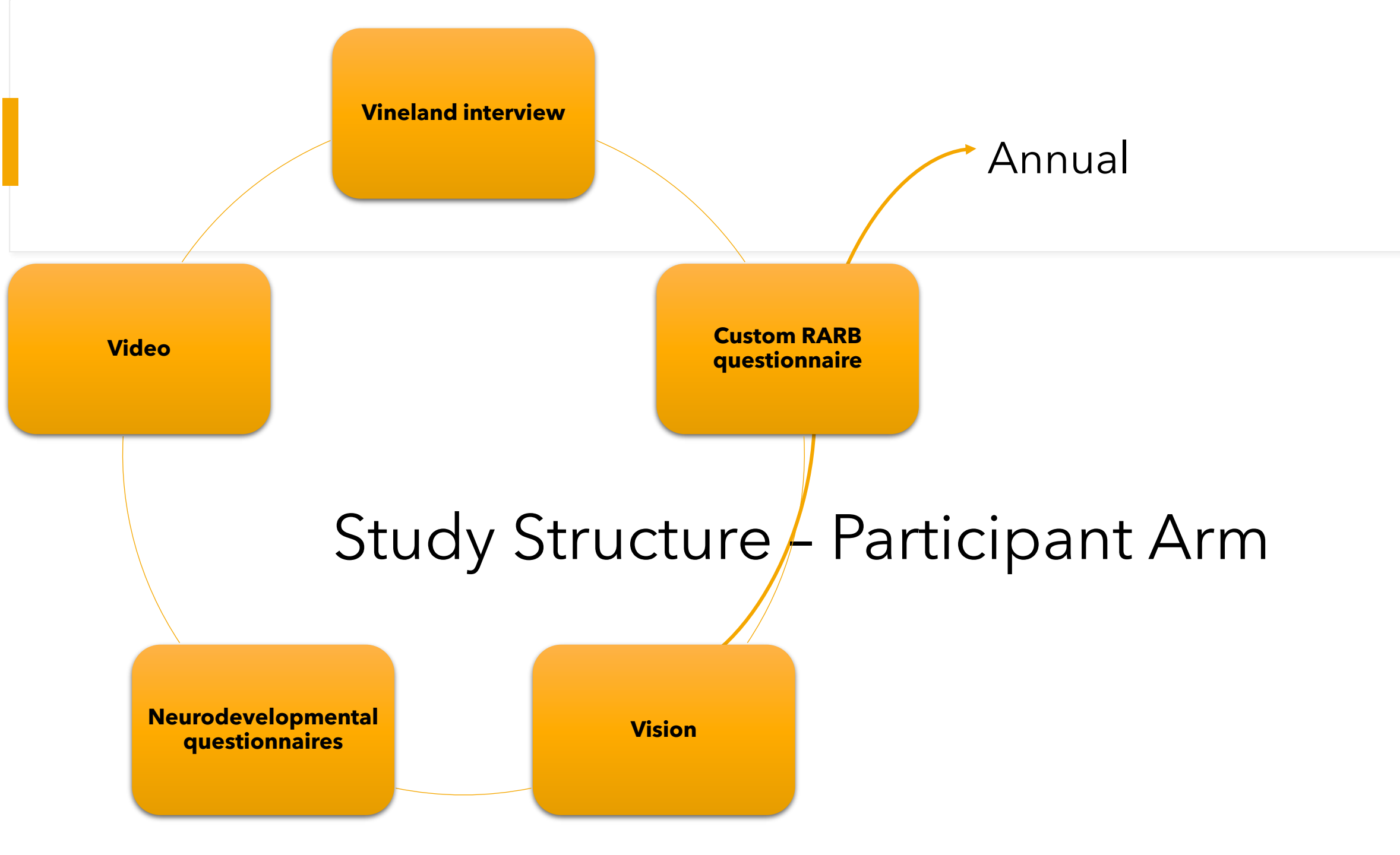
Video

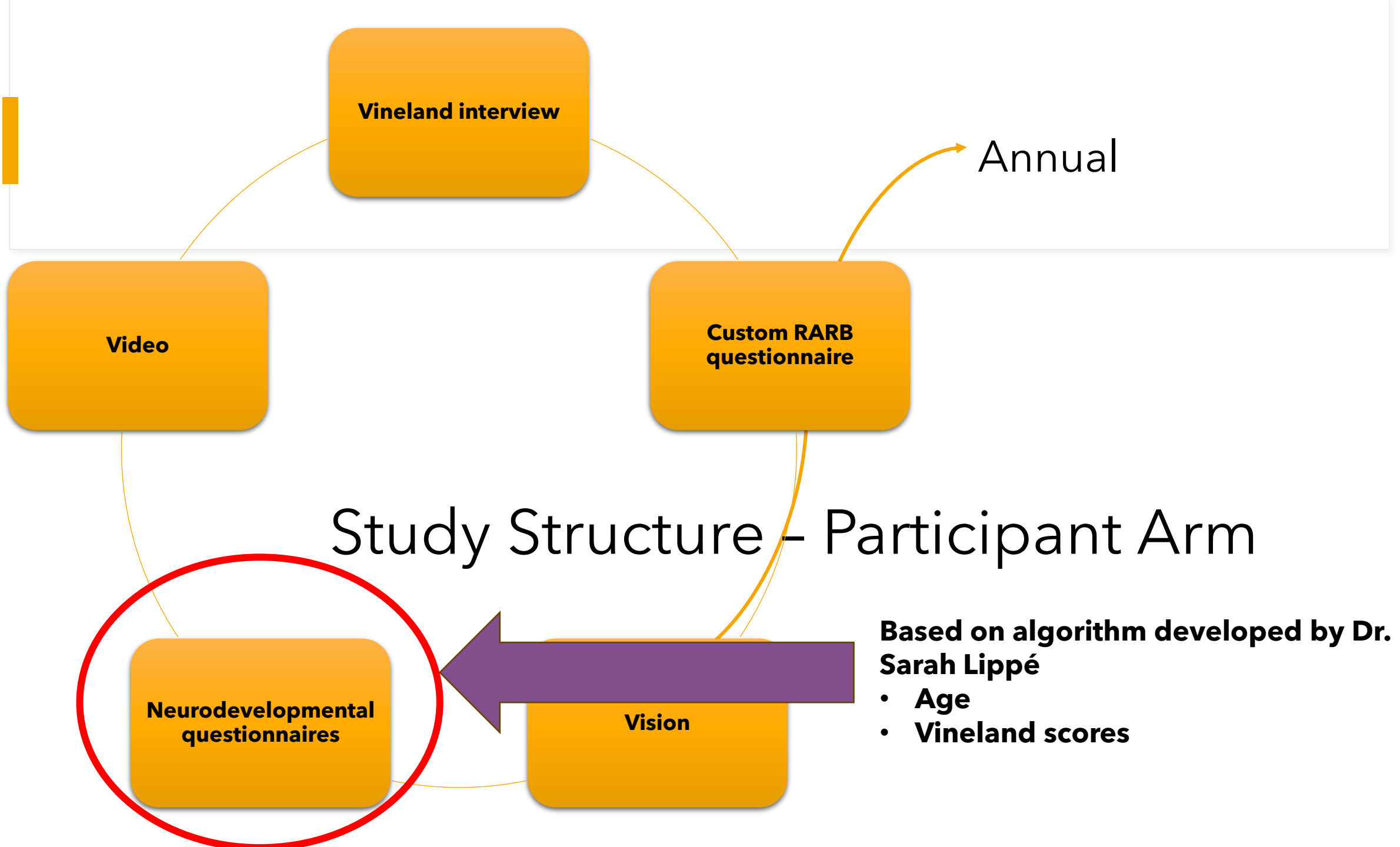
**Custom RARB
questionnaire**

Study Structure – Participant Arm

**Neurodevelopmental
questionnaires**

Vision





Study Structure – Physician Arm

One-time

Baseline data

As needed

Brain MRI

**Neurological
questionnaires**



Usually referring physician or
geneticist



Study Structure – Physician Arm

Baseline data

Brain MRI

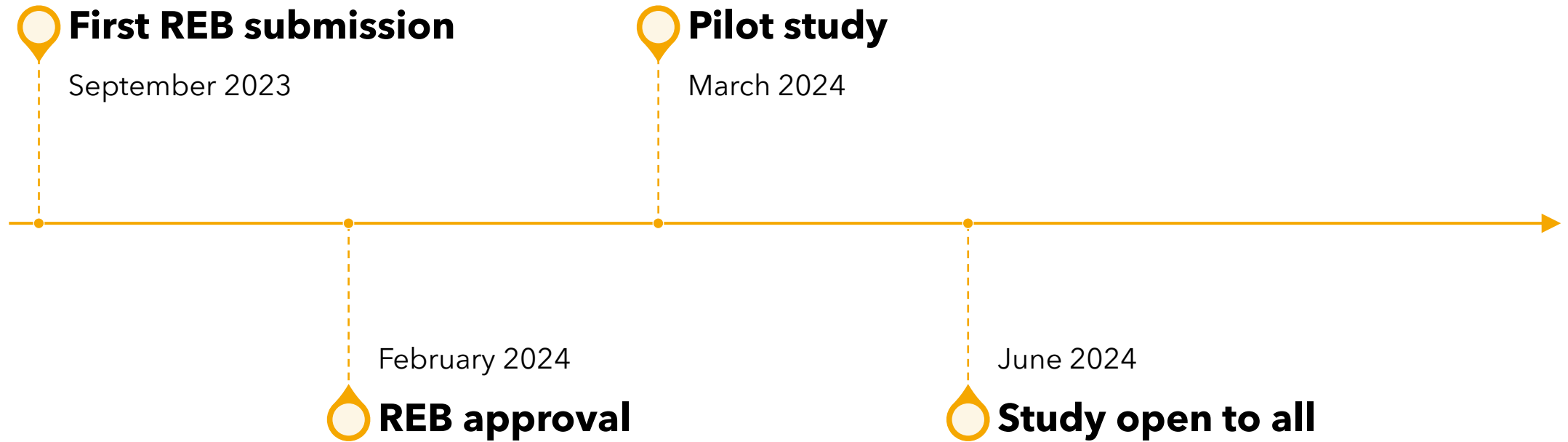
**Neurological
questionnaires**

Annual

Usually neurologist or
physiotherapist



A (Very Brief) Timeline





Lessons Learned so Far

- We rely on RARB families to push the study forward
- Please respond to our emails
- Follow up with your physicians to fill out:
 - Baseline genetic/clinical data
 - Annual neurological questionnaires